

Now assemble the master key. The versed sine gives $F \propto QR/t^2$, while the swept area sets $t \propto SP \cdot QT$; substitute and the force becomes reciprocally as the force-measure solid $SP^2 \cdot QT^2/QR$, taken in the limit as $Q \rightarrow P$ (Cor. 4 of Lem. X), and this solid is the heart of the whole method. For ANY curve, pick P and a nearby Q , draw $QR \parallel SP$ and $QT \perp SP$, and compute this limit — it is the single most used formula in all of Book I. Props. 7, 11, and 13 will feed the circle, the ellipse, and the parabola into exactly this construction. *Q.E.D.*